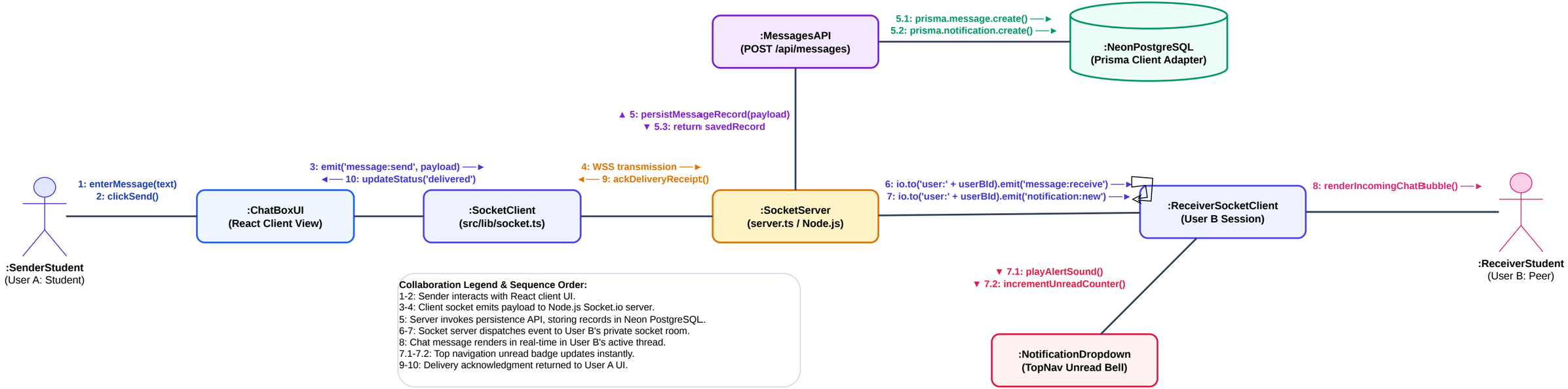
CampusConnect — UML Collaboration (Communication) Diagram

Real-Time Peer-to-Peer Messaging, Room Routing & Push Notification Broadcast Architecture



Collaboration Legend & Sequence Order:
1-2: Sender interacts with React client UI.
3-4: Client socket emits payload to Node.js Socket.io server.
5: Server invokes persistence API, storing records in Neon PostgreSQL.
6-7: Socket server dispatches event to User B's private socket room.
8: Chat message renders in real-time in User B's active thread.
7.1-7.2: Top navigation unread badge updates instantly.
9-10: Delivery acknowledgment returned to User A UI.